explanatory $\quad x_t \in \mathbb{R}^A \qquad t = 1, \cdots, T$

response $\quad y_t \in \mathbb{R}^B$

estimate $\quad \hat{y}_t \in \mathbb{R}^B \qquad$ with $\quad \hat{y}_t = \hat{f}(x_t)$

loss $\quad \mathcal{L}\left(\{y_t\}_{t=1}^{T}, \ \{\hat{y}_t\}_{t=1}^{T} \right) = \mathcal{L}(\underset{\text{weights}}{W}, \underset{\text{biases}}{b}) = \mathcal{L}$

regularizers:

$$\mathcal{L}^{reg} := \mathcal{L} + \|W\|_2^2 \qquad \text{ridge}$$

$$:= \mathcal{L} + \|W\|_1 \qquad \text{lasso}$$

$$:= \mathcal{L} + \alpha \|W\|_1 + (1-\alpha)\|W\|_2^2 \qquad \text{elastic net.}$$

$$\frac{\partial \mathcal{L}^{reg}}{\partial b_j^{(l)}} = \frac{\partial \mathcal{L}}{\partial b_j^{(l)}} \qquad \text{biases don't affect gradient.}$$

$$\frac{\partial \mathcal{L}^{reg}}{\partial W_{ij}^{(l)}} = \frac{\partial \mathcal{L}}{\partial W_{ij}^{(l)}} + \frac{\partial}{\partial W_{ij}^{(l)}} \left(\alpha \|W\|_1 + (1-\alpha)\|W\|_2^2 \right)$$

$$= \frac{\partial \mathcal{L}}{\partial W_{ij}^{(l)}} + \underbrace{\alpha \,\mathrm{sign}\left(W_{ij}^{(l)}\right) + 2(1-\alpha) \cdot W_{ij}^{(l)}}_{\text{just add these within Network::update\_parameters\_}}$$

calculated already.

Note: ignore the $\{\cdot\}_{t=1}^{T}$ notation was tackling time series cases when implementing regularisation.